Compression :-

- the process of reducing the size of data packets or files before transmission, using encoding techniques to minimize bandwidth usage and improve transfer speeds
- Compression works in the Presentation Layer of OSI Reference Model.

Overview of tar Compression Options

OPTION	DESCRIPTION
-z, --gzip	Use gzip compression (**.tar.gz**).
-j, --bzip2	Use bzip2 compression (**.tar.bz2**). bzip2 typically achieves a better compression ratio than gzip.
-J, --xz	Use xz compression (**.tar.xz**). The xz compression typically achieves a better compression ratio than bzip2.

Compression using gzip :-

```
[student@node1 thursday24]$ gzip doc1
[student@node1 thursday24]$ ls -l
total 12
-rw-r--r--. 1 student student 37 Oct 24 09:52 doc1.gz
-rw-r--r--. 1 student student 14 Oct 24 09:52 doc2
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc3
```

Decompression using gzip :-

```
[student@node1 thursday24]$ gunzip doc1.gz
[student@node1 thursday24]$
[student@node1 thursday24]$ ls -l
total 12
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc1
-rw-r--r--. 1 student student 14 Oct 24 09:52 doc2
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc3
[student@node1 thursday24]$
```

Note : in minimal installation some packages are not installed by default.
Before running any tool we must check if the package is installed or not .
To check a package is installed or not : $ rpm -qa <tool_name>

If the full package name is visible as the output that means the package is installed else no output will come.
We need to install the package using dnf / YUM : # dnf install <package>

Compression using bzip2 :-

```
[student@node1 thursday24]$ bzip2 doc2
[student@node1 thursday24]$ ls -l
total 12
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc1
-rw-r--r--. 1 student student 55 Oct 24 09:52 doc2.bz2
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc3
```

Decompression using bzip2 :-

```
[student@node1 thursday24]$ bunzip2 doc2.bz2
[student@node1 thursday24]$
[student@node1 thursday24]$ ls -l
total 12
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc1
-rw-r--r--. 1 student student 14 Oct 24 09:52 doc2
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc3
```

Compression using xz :-

```
[student@node1 thursday24]$ xz doc3
[student@node1 thursday24]$
[student@node1 thursday24]$ ls -l
total 12
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc1
-rw-r--r--. 1 student student 14 Oct 24 09:52 doc2
-rw-r--r--. 1 student student 68 Oct 24 09:52 doc3.xz
```

Decompression using xz :-

```
[student@node1 thursday24]$ unxz doc3.xz
[student@node1 thursday24]$
[student@node1 thursday24]$ ls -l
total 12
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc1
-rw-r--r--. 1 student student 14 Oct 24 09:52 doc2
-rw-r--r--. 1 student student 12 Oct 24 09:52 doc3
```

Tar :

THE tar COMMAND

Archiving and compressing files are useful when creating backups and transferring data across a network. One of the oldest and most common commands for creating and working with backup archives is the **`tar`** command.

With **`tar`**, users can gather large sets of files into a single file (archive). A **`tar`** archive is a structured sequence of file data mixed in with metadata about each file and an index so that individual files can be extracted. The archive can be compressed using `gzip`, `bzip2`, or `xz` compression.

The **`tar`** command can list the contents of archives or extract their files to the current system.

Overview of `tar` Operations

OPTION	DESCRIPTION
`-c`, `--create`	Create a new archive.
`-x`, `--extract`	Extract from an existing archive.
`-t`, `--list`	List the table of contents of an archive.

Selected tar General Options

OPTION	DESCRIPTION
-v, --verbose	Verbose. Shows which files get archived or extracted.
-f, --file=	File name. This option must be followed by the file name of the archive to use or create.
-p, --preserve-permissions	Preserve the permissions of files and directories when extracting an archive, without subtracting the umask.

TO create the archive :-

```
[student@node1 ~]$ mkdir result
[student@node1 ~]$
[student@node1 ~]$ mkdir pass
[student@node1 ~]$
[student@node1 ~]$ mkdir fail
[student@node1 ~]$
[student@node1 ~]$ touch result/file2
[student@node1 ~]$
[student@node1 ~]$ touch pass/file3
[student@node1 ~]$
[student@node1 ~]$ touch fail/file4
[student@node1 ~]$
[student@node1 ~]$ echo "Hello Linux" > data/file1
[student@node1 ~]$ echo "Hello windows" > result/file2
[student@node1 ~]$ echo "Hello cisco" > pass/file3
[student@node1 ~]$ echo "Hello Juniper" > fail/file4
[student@node1 ~]$
[student@node1 ~]$ tar -cvf files.tar data/file1 result/file2 pass/file3 fail/file4
data/file1
result/file2
pass/file3
fail/file4
[student@node1 ~]$
```

To view the content of the archive :-

```
[student@node1 ~]$ tar -tvf files.tar
-rw-r--r-- student/student  12 2024-10-24 10:38 data/file1
-rw-r--r-- student/student  14 2024-10-24 10:38 result/file2
-rw-r--r-- student/student  12 2024-10-24 10:38 pass/file3
-rw-r--r-- student/student  14 2024-10-24 10:38 fail/file4
[student@node1 ~]$
```

To extract to the original location :-

```
[student@node1 ~]$ tar -xvf files.tar
data/file1
result/file2
pass/file3
fail/file4
[student@node1 ~]$ ls
```

To extract file on a new location :-

```
[student@node1 ~]$ mkdir newfolder
[student@node1 ~]$
[student@node1 ~]$ tar -xvf files.tar -C newfolder/
```

Compressing and archiving together

Overview of tar Compression Options

OPTION	DESCRIPTION
-z, --gzip	Use gzip compression (**.tar.gz**).
-j, --bzip2	Use bzip2 compression (**.tar.bz2**). bzip2 typically achieves a better compression ratio than gzip.
-J, --xz	Use xz compression (**.tar.xz**). The xz compression typically achieves a better compression ratio than bzip2.

Compression and archive using gzip & tar

```
[student@node1 ~]$ tar -zcvf compressed_backup.tar.gz data/file1 result/file2 pass/file3 fail/file4
data/file1
result/file2
pass/file3
fail/file4
[student@node1 ~]$
[student@node1 ~]$ ls
asmit                    data   database  fail   file5      newfolder  result  thursday24
compressed_backup.tar.gz data2  exam      file4  files.tar  pass       test
[student@node1 ~]$
```

Compression and archive using bzip2 & tar

```
[student@node1 ~]$ tar -jcvf compressed_backup.tar.bz2 data/file1 result/file2 pass/file3 fail/file4
data/file1
result/file2
pass/file3
fail/file4
[student@node1 ~]$ ls
asmit                      compressed_backup.tar.gz  data2     exam  file4  files.tar  pass     test
compressed_backup.tar.bz2  data                      database  fail  file5  newfolder  result   thursday24
[student@node1 ~]$
```

Compression and archive using xz & tar

```
[student@node1 ~]$ tar -Jcvf compressed_backup.tar.xz data/file1 result/file2 pass/file3 fail/file4
data/file1
result/file2
pass/file3
fail/file4
[student@node1 ~]$ LS
-bash: LS: command not found
[student@node1 ~]$ ls
asmit                      compressed_backup.tar.gz  data   database  fail   file5      newfolder  result  thursday24
compressed_backup.tar.bz2  compressed_backup.tar.xz  data2  exam      file4  files.tar  pass       test
```

TRANSFERRING FILES USING SECURE COPY

OpenSSH is useful for securely running shell commands on remote systems. The Secure Copy command, **scp**, which is part of the OpenSSH suite, copies files from a remote system to the local system or from the local system to a remote system. The command uses the SSH server for authentication and encrypts data when it is being transferred.

You can specify a remote location for the source or destination of the files you are copying. The format of the remote location should be in the form **`[user@]host:/path`**. The **`user@`** portion of the argument is optional. If it is missing, your current local username will be used. When you run the command, your **scp** client will authenticate to the remote SSH server just like **ssh**, using key-based authentication or prompting you for your password.

TRANSFERRING FILES BETWEEN SYSTEMS SECURELY

```
[student@node1 ~]$ scp newbackup1.tar.bz2 student@192.168.56.110:/home/student/
student@192.168.56.110's password:
newbackup1.tar.bz2                                            100%  241   111.1KB/s   00:00
[student@node1 ~]$
```

TRANSFERRING FILES USING THE SECURE FILE TRANSFER PROGRAM

To interactively upload or download files from a SSH server, use the Secure File Transfer Program, **`sftp`**. A session with the **`sftp`** command uses the secure authentication mechanism and encrypted data transfer to and from the SSH server.

Just like the **`scp`** command, the **`sftp`** command uses **`[user@]host`** to identify the target system and user name. If you do not specify a user, the command will attempt to log in using your local user name as the remote user name. You will then be presented with an `sftp>` prompt.

```
[student@node1 ~]$ sftp student@192.168.56.110
student@192.168.56.110's password:
Connected to 192.168.56.110.
sftp>
sftp>
sftp> put newbac
newbackup.tar.gz    newbackup1.tar.bz2  newbackup2.tar.xz
sftp> put newbackup2.tar.xz
Uploading newbackup2.tar.xz to /home/student/newbackup2.tar.xz
newbackup2.tar.xz                                             100%  276    94.7KB/s   00:00
sftp>
sftp>
sftp>
sftp> get akash
Fetching /home/student/akash to akash
akash                                                         100%   41    11.6KB/s   00:00
sftp>
sftp> ls
Desktop             Documents           Downloads           Music               Pictures            Public
Templates           Videos              akash               newbackup.tar.gz    newbackup1.tar.bz2  newbackup2.tar.xz
sftp>
```